

JOONWON LEE

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Goldman Sachs
Global Banking & Markets
Global Currency and Emerging Markets
New York, NY

Re: Quantitative Strategist, Global Currency and Emerging Markets (GCEM)

Dear GCEM Hiring Team,

I am applying for the Quantitative Strategist position within Goldman Sachs' Global Currency and Emerging Markets team. I am a Statistics Ph.D. candidate at Rutgers University and expect to complete my degree in December 2026. My training in stochastic modeling, machine learning, numerical optimization, and simulation, together with experience in quantitative finance, insurance pricing, and algorithmic market making, aligns with a desk-strat role combining quantitative research, production-oriented coding, and close collaboration with traders.

At JPMorgan Chase, I developed and validated a Sequential Probability Ratio Test framework for binary risk indicators. I translated a monitoring problem into statistically controlled decision rules and used simulation to evaluate Type I and Type II error, detection delay, and early-decision tradeoffs. Brownian-motion approximations characterized stopping-time uncertainty and helped me communicate the model's behavior and limitations.

My doctoral research develops scalable approximate-likelihood inference and statistical diagnostics for large, dependent datasets. I built an advection-aware Vecchia approximation for nonseparable Gaussian processes, replacing dense covariance operations with fixed-budget conditioning to fit up to 145,008 satellite observations per day. The Python/PyTorch implementation uses GLS profiling, L-BFGS optimization, CPU/GPU execution, and restartable HPC workflows. Although the application is atmospheric data, the work has trained me to calibrate and test models under dependence, missingness, noise, and computational constraints.

At Travelers, I developed an end-to-end LightGBM pricing pipeline across more than 2.46 million property-risk records, benchmarked it against a generalized linear model, and presented performance and feature-attribution findings as pricing and risk-segmentation recommendations. I also placed in the top 2.39% of the IMC Prosperity Algorithmic Trading Competition by designing a market-making strategy with inventory-aware quoting, dynamic liquidation rules, and benchmark-driven evaluation.

I would welcome the opportunity to bring this combination of statistical depth, practical Python engineering, and model-validation discipline to GCEM. I am eager to deepen my product knowledge across Rates, FX, and Credit while contributing to machine-learning research and robust trading tools. Thank you for your consideration.

Sincerely,
Joonwon Lee